The features of a bishop: Gender, number, and Impoverishment in BCS

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NELS 57: Oct 16-18, 2026

1. Introduction

- Bosnian/Croatian/Serbian (BCS) has a set of nouns with an unusual gender agreement pattern:
  - Uniform masculine agreement in the singular, (1-a)
  - Variable feminine or masculine agreement in the plural, (1-b)[1]

(1) a. Nov-**i**/*-**a** vladika je sazva-**o**/*-**la** sastanak. **M Sg Agr**
new-**M.SG**/*-**F.SG** bishop AUX called-**M.SG**/*-**F.SG** meeting
'The new bishop called a meeting.'

b. Nov-**i**/-**e** vladike su sazval-**i**/-**e** sastanak. **M/F Pl Agr**
new-**M.PL/F.PL** bishops AUX.PL called-**M.PL/F.PL** meeting
'The new bishops called a meeting.'

→ This set of nouns includes the noun *bishop* and we refer to them henceforth as *bishop* nouns[2]

- *Bishop* nouns have attracted significant attention in the morphosyntactic literature (see Wechsler & Zlatić 2000, 2003, Corbett 2010, 2015, Despić 2017, Puškar 2018, Franks 2020, i.a.)
  - This work has yielded substantial insights, but many of the analyses require significant analytical machinery and sets of interlocked stipulations unique to *bishop* nouns
  - Almost all previous analyses assume that *bishop* nouns are *hybrid* nouns, i.e., nouns that have both social and grammatical gender features

*For helpful discussion of different aspects of this work, we are grateful to audiences at the Boston Morphology Workshop, FASL 35, WCCFL 44 and the Syntax Reading Group at Georgetown. Thanks are also due to Lefteris Paparounas, Tanja Milićev, Masha Polinsky, Luis Miguel Toquero-Pérez, Matthew Hewett, and Mal Shah.

[1] All data points without a citation are from the judgments of the first author unless otherwise stated.

[2] These nouns are also sometimes called *split hybrid* nouns because of their distinct agreement behavior in the singular vs. plural (see e.g., Corbett 1991, 2015, Puškar 2018).

- We develop a new analysis of *bishop* nouns, arguing that they have both [+FEM] and [+MASC] grammatical gender features:
  - In the singular, a **markedness-targeting** Impoverishment operation targets and deletes [+FEM], resulting in masculine agreement (see e.g., Noyer 1992, 1997, 2001, Nevins 2011, Arregi & Nevins 2012; see also Despić 2017 on BCS)
  - In the plural, a **markedness-triggered** Impoverishment operation deletes **either** [+MASC] **or** [+FEM], resulting in variability (see e.g., Bobaljik 2001, Nevins 2011, Arregi & Nevins 2012, Kramer 2016, 2019)
- We argue that this analysis is both simpler than previous approaches and has better empirical coverage in BCS
  - We bring to bear new evidence that *bishop* nouns are **not** in fact hybrid nouns, and the Impoverishment analysis allows them to have grammatical gender features only
  - The analysis also addresses a long-standing puzzle about the nature of gender features in three-gender languages (see e.g., Kramer 2015 vs. Despić 2016)
  - Finally, the analysis successfully accounts for the behavior of *bishop* nouns in certain additional Slavic languages, providing further support for an Impoverishment approach

Roadmap

- Section 2: Two puzzles: *bishop* nouns and the nature of gender features
- Section 3: The proposal: Impoverishment operations
- Section 4: Empirical ramifications and theoretical implications (incl. the evidence that *bishop* nouns are not hybrid)
- Section 5: Cross-Slavic predictions (*bishop* nouns in Russian, Slovenian, and Old Church Slavonic)
- Section 6: Conclusion

2. The two puzzles

2.1. *Bishop nouns*

- In BCS, a relatively small subset of male-denoting nouns display the variable agreement pattern seen in (1): masculine in singular, masculine or feminine in plural
- *bishop* nouns include *tata* 'dad', *deda*, 'grandpa', *papa* 'pope', *vojvoda* 'duke', *kolega* 'colleague', *gazda* 'owner', *sluga* 'servant', *komšija* 'neighbor', *bekrija* 'boozer', etc.

(2) a. Naš-**ø**/*-**a** vojvoda je bi-**o**/*-**la** u pravu. **M Sg Agr**
our-**M.SG**/*-**F.SG** duke AUX was-**M.SG**/*-**F.SG** in right
'Our duke was in the right.'

b. Naš-**i**/-**e** vojvode su bil-**i**/-**e** u pravu. **M/F Pl Agr**
our-**M.PL/F.PL** dukes AUX.PL were-**M.PL/F.PL** in right
'Our dukes were in the right.'

- *Bishop* nouns are in Declension Cass II, which mostly includes nouns with feminine grammatical gender (whereas masculine nouns are primarily found in class I)

(3) *BCS noun classes* (adapted from Mrazović & Vukadinović 1990 via Puškar 2018)

Class	Example	Ending	Gender
I_M	*otac*-ø 'father', *krov*-ø 'roof', *aut-o* 'car', *Đorđ-e* 'George'	*-ø*, *-o*, *-e*	M
I_N	*sel-o* 'village', *mor-e* 'sea'	*-o*, *-e*	N
II	*majk-a* 'mother', *kuć-a* 'house'	*-a*	F
	***vladik-a* 'bishop', *vojvod-a* 'duke'**		?
III	*ljubav*-ø 'love', *čast*-ø 'honor'	*-ø*	F

- Class II membership and variable agreement behavior in the plural have formerly been taken as evidence that *bishop* nouns are grammatically feminine (with masculine social gender)
  - This would fold them into a more familiar phenomenon: *hybrid* nouns
  - A typical hybrid noun like *pedagog* 'educator' triggers masculine agreement when it refers to a man or woman, and feminine agreement when it refers to a woman (4)
  - This is usually taken to mean that the noun *pedagog* 'educator' has masculine grammatical gender and optional feminine social gender

(4) a. Pedagog nas je pozva-**o** na telefon. **(male or female)**
educator us AUX called-**M.SG** on phone
'The educator called us on the phone.'

b. Pedagog nas je pozva-**la** na telefon. **(female only)**
educator us AUX called-**F.SG** on phone
'The educator called us on the phone.'

- However, *bishop* nouns do not behave like typical hybrid nouns, since agreement with hybrid nouns is not sensitive to number (see also Section 4)
- Capturing the variable agreement behavior of *bishop* nouns has so far required significant analytical machinery and many assumptions specific to *bishop* nouns (again, see Section 4)

2.2. *Gender representation in three-gender languages*

- It is controversial how to properly characterize grammatical gender features in languages like BCS, which have a three-way gender system
- Some previous work has argued for (5), where each gender is identified by two binary gender features: [+/-MASC] and [+/-FEM] (e.g., Despić 2016, Opitz & Pechmann 2016, Despić & Murray 2018, Privizentseva 2023, Matushansky 2025, i.a.)

(5) a. [+FEM][-MASC] = feminine
b. [-FEM][+MASC] = masculine
c. [-FEM][-MASC] = neuter
d. [+FEM][+MASC] = ??

- As (5-d) indicates, one part of the typology is missing: it is said that [+FEM][+MASC] is unattested because it would lead to a feature conflict (e.g., Matushansky 2025:439)
- However, the explanation for the absence of (5-d) is odd:
  - The tacit assumption might be that there is some kind of semantic 'conflict' between being both male and female...?
  - But in these approaches, (5) is also assumed to hold for *grammatical* gender features, in addition to (and/or separately from) social gender
  - If there is some kind of conflict when both grammatical gender features have the same value, then neuter should be a problem as well, since it is doubly specified with [-]

Our contribution

- The challenging gender/number agreement pattern of *bishop* nouns, and the mysterious empty cell in the typology in (5), can be solved at the same time!
- The key idea: *bishop* nouns have both [+FEM] and [+MASC] features = (5-d)
- The gender/number agreement patterns emerge from two familiar kinds of Impoverishment rules, which work in tandem in BCS

- Next section: we lay out the details of how to Impoverish a (singular or plural) bishop

3. The proposal: Feature specification and Impoverishment

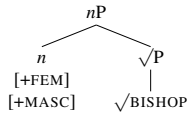
- Road map for this section
  - 3.1 The (gender) features of a bishop
  - 3.2 Bishop in the singular: markedness-targeting Impoverishment results in M agr
  - 3.3 Bishop in the plural: markedness-triggered Impoverishment results in M/F agr

3.1. *The (gender) features of a bishop*

- BCS has the gender system in (6):

(6) a. [+FEM][-MASC] = feminine, e.g., *majka* 'mother,' *kuća* 'house'...
b. [-FEM][+MASC] = masculine, e.g., *otac* 'father,' *krov* 'roof'...
c. [-FEM][-MASC] = neuter, e.g., *selo* 'village,' *more* 'sea'...
d. **[+FEM][+MASC], e.g., *vladika* 'bishop,' *kolega* 'colleague'...**

- Hence, bishop nouns have [+MASC] [+FEM] *grammatical gender*, which is encoded on *n*:

(7)


= *vladika* 'bishop'

- (6) allows for a straightforward analysis of gender and declension class in BCS
- We assume declension class is assigned via rule at PF (see e.g., Oltra-Massuet 1999, Embick & Halle 2005, Kramer 2015); such rules frequently reference grammatical gender
- Only three ordered rules are required for BCS declension classes:

(8) a. [list of roots] → [III] (e.g., *stvar* 'thing', *ljubav* 'love', *smrt* 'death')
b. [+FEM] → [II] (e.g., *Marija* 'Mary', *lopta* 'ball', *vladika* 'bishop')
c. default → [I], e.g., (e.g., *Petar* 'Peter', *auto* 'car', *more* 'sea')

  - Class III is assigned based on root identity
  - Class II is assigned to nouns with [+FEM], including *bishop* nouns
  - Class I is assigned by default to the remaining nouns (all masculine or neuter)

- Given the feature specification in (6) and the rules in (8) we successfully group *bishop* nouns in Class II and correctly assign declension class to the remaining nominals in BCS
- Declension class rules like (8) are attested in other languages, e.g., Spanish (Kramer 2015)
- (6) also allows for uncontroversial assumptions about the markedness of gender features:

(9) a. The [+] value is marked; the [-] value is unmarked.
b. [+FEM] is marked relative to [+MASC]. (see e.g., Jakobson 1984/1932, Bale et al. 2011, Despić 2017)

- (9) is the foundation for our Impoverishment analysis of gender agreement with *bishop* nouns

3.2. *Bishop in the singular: markedness-targeting Impoverishment*

- Impoverishment is a postsyntactic operation that deletes morphosyntactic features (originally proposed in Bonet 1991, see Keine & Müller to appear for a recent overview)
  - It occurs before Vocabulary Insertion
  - Because it deletes features, it leads to the insertion of less-specified exponents than indicated by the syntactic/semantic context ("retreat to the default")
- One major motivation for Impoverishment is markedness, in two distinct senses (see e.g., Noyer 1992, 1997, 2001, Nevins 2011, Bobaljik 2001, Harley 2008, Arregi & Nevins 2012):
  - An excess of markedness in a feature bundle causes a marked feature to delete = markedness-*targeting* Impoverishment (this section)
  - An excess of markedness in a feature bundle causes some unrelated feature(s) to delete = markedness-*triggered* Impoverishment (Section 3.3)
  - The difference is "whether a deletion operation whose structural description mentions a marked feature-value [aF] in fact deletes [aF] itself (markedness-targeted), or whether it deletes an orthogonal feature to [F], e.g. [bG]." (Arregi & Nevins 2012:194)
- Example of markedness-targeting Impoverishment: dual in Classical Arabic
  - We assume the number features [+/-PLURAL] and [+/-MINIMAL] (cf. Harbour 2020[3])

(10) a. [-PL][+MIN] = singular
b. [+PL][-MIN] = plural
c. [+PL][+MIN] = dual
d. [-PL][-MIN] = impossible (Harbour, 2020, 66)

  - Dual is marked with respect to plural (see e.g., Nevins 2011): in (10), it has the plus value for [MIN]
  - In Classical Arabic, first person dual is always realized in the same way as first person plural (Hewett, 2023, 152)
  - Example: jussive prefix conjugation in (11)

(11) *Classical Arabic jussive prefix conjugation (Hewett, 2023, 152, (40))*

	SG	DU	PL
1	**ʔa**-fʕal	**na**-fʕal	**na**-fʕal
2M	**ta**-fʕal	**ta**-fʕal-**a:**	**ta**-fʕal-**u:**
2F	**ta**-fʕal-**i:**	**ta**-fʕal-**a:**	**ta**-fʕal-**na:**
3M	**ya**-fʕal	**ya**-fʕal-**a:**	**ya**-fʕal-**u:**
3F	**ta**-fʕal	**ta**-fʕal-**a:**	**ya**-fʕal-**na**

[3]We use [+/-PLURAL] in the place of [+/-ATOMIC] for ease of exposition.

– Analysis: a markedness-targeting Impoverishment operation deletes the marked[+MIN] feature in non-singular 1st person contexts

(12) **Markedness-Targeting Impoverishment for Dual in Classical Arabic**
a. SD: an Agr node specified as [+AUTHOR][+PL][+MIN]
b. SC: Delete [+MIN] in Agr (modified from Hewett, 2023, 153, (42))

– Result: only [+PL] remains available to determine insertion, so plural forms are predicted to be inserted[4]

- **Key proposal for BCS singular *bishop* nouns:** elements that agree with singular *bishop* nouns undergo markedness-targeting Impoverishment

– *Bishop* nouns have two marked gender features, [+MASC] and [+FEM], which is the most marked combination of gender features as per (9)

– Impoverishment deletes the most marked feature of these two: [+FEM]

(13) **Markedness-Targeting Impoverishment for Agr with *bishop* Nouns**
a. SD: an Agr node specified as [+MASC][+FEM]
b. SC: Delete [+FEM] in Agr

– Result: only [+MASC] remains, so obligatory masculine agreement is predicted with *bishop* nouns, as in (14-a)[5]

(14) a. Nov-**i/*-a** vladika je sazva-**o/*-la** sastanak. **M SgAgr**
new-**M.SG/*F.SG** bishop AUX called-**M.SG/*F.SG** meeting
'The new bishop called a meeting.'

b. Nov-**i/-e** vladike su sazval-**i/-e** sastanak.**M/F PlAgr**
new-**M.PL/F.PL** bishops AUX.PL called-**M.PL/F.PL** meeting
'The new bishops called a meeting.'

- But what about the plural nouns, which trigger variable agreement, (14-b)? Next up!

3.3. *Bishop in the plural: markedness-triggered Impoverishment*

- Since Greenberg 1966, it has been recognized that languages neutralize gender distinctions in the plural, i.e., languages have 'fewer' genders in the plural

– For example, Coptic (Egyptian (Afroasiatic)) does not distinguish between masculine and feminine gender in the plural (Kramer 2016, 2019)

– Example: definite determiners in (15)

[4]Given certain plausible assumptions about the featural content of Vocabulary Items in Classical Arabic

[5]Since [+FEM] is deleted, then the relevant masculine agreement Vocabulary Items in BCS must also lack [-FEM]; otherwise, the Subset Principle would prevent any Vocabulary Item from being inserted. As far as we can determine, nothing negative follows from this. Alternatively, Impoverishment might delete the marked value for [FEM] and then insert the unmarked value (-) (Noyer 2001, Harbour 2003), which nullifies the issue.

(15) *Coptic Definite Determiners (Kramer, 2016, 100)*

	FEMININE	MASCULINE
SINGULAR	tə-	pə-
PLURAL	nə-	

– This generalization holds in Coptic across all feature bundles that show phi-agreement in a broad sense (agreement inflection, pronouns, etc.)

- Major line of analysis for this kind of generalization: markedness-triggered Impoverishment

– Coptic has two genders and two numbers so we assume simple gender/number features: [+/-FEM] and [+/-PL]

– Gender features are deleted from feature bundles that include a plural feature

(16) **Markedness-Triggered Impoverishment for Gender in Coptic**
a. SD: an Agr node specified as [+PL]
b. SC: Delete [+/-FEM] in Agr (modified from Kramer, 2016, 102, (7))

– Importantly, the plural feature itself is not deleted: the unrelated gender features are deleted due to the markedness of the plural

- BCS has an Impoverishment operation like (16), that deletes all gender features in certain plural contexts (see e.g., Despić, 2017, 272)

– For example, gender distinctions are neutralized in plural relative pronouns with oblique case (across Vocabulary Items)

(17) *Plural Relative Pronouns in BCS (Arsenijević & Gračanin-Yuksek, 2016, 7)*

	MASCULINE		FEMININE		NEUTER	
NOM	koj-i which-M.PL	stolovi tables.M	koj-e which-F.PL	jabuke apples.F	koj-a which-N.PL	sunca suns.N.PL
ACC	koj-e	stolove	koj-e	jabuke	koj-a	sunca
GEN	koj-**ih**	stolova	koj-**ih**	jabuka	koj-**ih**	sunaca
INST	koj-**im**	stolovima	koj-**im**	jabukama	koj-**im**	suncima
DAT/ LOC	koj-**im**	stolovima	koj-**im**	jabukama	koj-**im**	suncima

(18) **Markedness-Triggered Impoverishment for Both Gender Features in BCS**[6]
a. SD: An oblique-case relative pronoun specified as [+PL]
b. SC: Delete [+/-FEM] and [+/-MASC] in the pronoun

[6]Similar patterns are found in other plural contexts (e.g., non-nominative personal pronouns), and we assume these contexts are listed out in the full SD for (18).

– If all the gender features are deleted, as in (18), then *bishop* nouns are predicted to trigger the same relative pronoun agreement as any other plural noun - borne out!

(19) Vladike koj-**im** sam dao veliku donaciju su me zapamtile.
bishops who-**DAT.PL** AUX gave big donation AUX.PL me remembered
'The bishops to whom I gave a big donation remembered me.'

– However, BCS does **not** neutralize genders in other plural contexts

(20) a. Ov-**i** stolovi su bil-**i** u hodniku.
these-**M.PL.NOM** tables.M AUX were.PTCP-**M.PL** in hallway
'These tables were in the hallway.'

b. Ov-**e** jabuke su bil-**e** na suncu.
these-**F.PL.NOM** apples.F AUX were.PTCP-**F.PL** in sun
'These apples were in the sun.'

c. Ov-**a** sunca su bil-**a** u našoj galaksiji.
these-**N.PL.NOM** suns.N AUX were.PTCP-**N.PL** in our galaxy
'These suns were in our galaxy.'

– So, in certain contexts, BCS tolerates the co-occurrence of [+FEM] with a plural feature as well as the co-occurrence of [+MASC] with a plural feature

– However, it is in these contexts that the *bishop* nouns have variable agreement in gender

(21) Ov-**i/-e** vladike su bil-**i/-e** u Pećkoj patrijašiji.
these-**M.PL/F.PL.NOM** bishops AUX were-**M.PL/F.PL** in Peć patriarchate
'These bishops were in the Patriarchate of Peć.'

∗ We conclude that it is the **combination** of both [+FEM] and [+MASC] with [+PL] that the grammar cannnot tolerate in these contexts

- **Key proposal:** elements that agree with plural *bishop* nouns undergo markedness-triggered Impoverishment

(22) **Markedness-Triggered Impoverishment for Agr with *bishop* Nouns**
a. SD: an Agr node specified as [+MASC][+FEM][+PL]
b. SC: Delete [+FEM] or [+MASC] in Agr

– If [+MASC] is deleted, then the result is feminine agreement

– If [+FEM] is deleted, then the result is masculine agreement

– Correct prediction = variable agreement

- Question: why aren't both gender features deleted in (22)?

– For economy reasons: this context is not as marked as the oblique-case plural context so there is no need to delete as many features to lower the markedness threshold (tentative)

– Alternatively, deleting both gender features is an option **alongside** deleting the individual gender features

∗ If both features are deleted, the result would be a feature bundle with just [+PL]

∗ It may be that a [+PL] feature bundle is realized with a 'masculine' agreement exponent (TBD if this is plausible in BCS, depending on defaults)

- What is the relationship between the markedness-targeting Impoverishment operation in (13), repeated in (23), and the markedness-triggering Impoverishment operation in (22)?[7]

– In particular, both operations could in principle apply to the feature bundle [+MASC][+FEM][+PL]

(23) **Markedness-Targeting Impoverishment for Agr with *bishop* Nouns**
a. SD: an Agr node specified as [+MASC][+FEM]
b. SC: Delete [+FEM] in Agr

– However, we assume that Impoverishment operations are subject to the typical constraints on PF operations, e.g., the Elsewhere Condition (see Arregi & Nevins 2012)

– (22) is more specific than (23) since its SD includes [+PL] ⇒ (22) applies before (23)

– (22) bleeds (23) because (22) deletes a gender feature

Interim summary

- *bishop* nouns have [+FEM] *and* [+MASC] features—the most marked gender in BCS!
- In the singular, a markedness-targeting Impoverishment operation deletes the most marked feature of them all [+FEM] = masc agreement
- In the plural, a markedness-triggered Impoverishment operation deletes either gender feature = masc/fem agreement

4. Empirical ramifications and theoretical implications

- Some broader consequences of an Impoverishment approach to *bishop* nouns:

– 4.1 Agreement resolution with coordinated plurals (evidence that resolution precedes Impoverishment)

– 4.2 Gender variation as an intra-speaker phenomenon (and initial evidence against treating *bishop* nouns as hybrid nouns)

– 4.3 [+MASC] on *bishop* nouns behaves like grammatical gender, not social gender (further evidence that *bishop* nouns are not hybrid nouns)

– 4.4 *bishop* nouns in BCS do not require special treatment (comparison to previous approaches)

[7] Many thanks to Neil Myler for discussion on this point.

4.1. *Agreement resolution with coordinated plurals*

- In BCS, if any number of plural animate nouns are coordinated, masculine gender agreement will surface if any of the conjuncts are masculine:

  (24) Student-kinj-e i profesor-i su bil-**i** ispred učionice.
  student-F-PL and professor.M-PL AUX.PL were-**M.PL** in-front classroom
  'The female students and the professors were in front of the classroom.'

- Feminine and neuter agreement surface only if all conjuncts have feminine or neuter gender (e.g., Puškar 2013)

  (25) Student-kinj-e i profesor-k-e su bil-**e** u prvim redovima.
  student-F-PL and professor-F-PL AUX.PL were-**F.PL** in first rows
  'The female students and the female professors were in the first rows.'

- Our analysis correctly predicts that only masculine agreement is possible when plural *bishop* nouns are coordinated with plural feminine nouns

  (26) Monahinje i vladike su sazval-**i/*-e** sastanak.
  nuns.F and bishops AUX.PL called-**M.PL/*F.PL** meeting
  'The nuns and the bishops called a meeting.'

- Since *bishop* nouns have both [+FEM] and [+MASC], the input to the coordination resolution mechanism in (26) will be the same as in the three-member mixed coordination in (27)

  (27) Monahinje, sveštenici i igumanije su sazval-**i/*-e** sastanak.
  nuns.F priests.M and abbesses.F AUX.PL called-**M.PL/*F.PL** meeting
  'The nuns, priests, and abbesses called a meeting.'

- Given the presence of [+MASC], masculine agreement will always win out in coordination resolution with plural *bishop* nouns, including coordinations of multiple plural *bishop* nouns:

  (28) Pape i vladike su oduvek živel-**i/***-**e** u raskoši.
  popes and bishops AUX.PL always lived-**M.PL/*F.PL** in luxury
  'Popes and bishops have always lived in luxury.'

- If our analysis of *bishop* nouns is correct, (26)/(28) argues for a broader theoretical point:
  - ⋆ **Agreement resolution happens before Impoverishment**
- If Impoverishment preceded resolution, [+MASC] could be deleted from the agreement target in (26)/(28), resulting in two [+FEM] features, leading to feminine agr—contrary to fact
- Coordination resolution is then firmly within the purview of the grammar (contra e.g., Lyskawa 2021, 2024, who argues that resolution is *extra-grammatical*)

4.2. *Gender variation as an intra-speaker phenomenon*

- Our approach captures the fact that variable gender agreement in the plural is an intra-speaker phenomenon (contra proposals that treat it as inter-speaker variation, e.g., Despić 2017)
- The Web corpus (SrWaC) contains many examples where agreement with *bishop* nouns is variable in the same text and even in the same sentence
- (29) illustrates mixed agreement with VP and clausal coordination; (30) with relative clauses; (31) with attributive vs. predicative agreement
- (32) illustrates (exceedingly rare) mixed agreement with multiple attributive adjectives

(29) a. Pape su krunisal-**e** germanske varvarske osvajače i
popes AUX.PL crowned-**F.PL** Germanic barbarian conquerors and
proglašaval-**i** ih vladarima Rimskog carstva.
declared-**M.PL** CL emperors Roman empire
'The popes crowned Germanic barbarian conquerors and declared them rulers of the Roman empire.'

b. Pape su veroval-**e** da je car Lav III jeretik kada je
popes AUX.PL believed-**F.PL** that AUX tsar Leo III heretic when AUX
zabranio ikone i počel-**i** su da sumnjaju...
forbade icons and started-**M.PL** AUX.PL DA doubt
'The popes believed that emperor Leo III was a heretic when he forbade icons and they started to doubt...'

(30) a. Nego se pitam šta traže pojedin-**e** vladike koj-**i** prošle godine
but SE wonder what seek some-**F.PL** bishops who-**M.PL** last year
nisu došl-**i** da daju podršku pravoslavnom narodu...
didn't come-**M.PL** to give support Orthodox people
'But I'm wondering what some bishops are seeking who last year did not come to support the Orthodox people...'

b. Od tada su pravoslavn-**e** vladike koj-**i** su protiv posjete
from then AUX.PL Orthodox-**F.PL** bishops who-**M.PL** AUX.PL against visit
pontifeka progonjen-**e** od propapskih državnih i crkvenih kadrova.
pontifex persecuted-**F.PL** by pro-papal state and church personnel
'From then on, Orthodox bishops who were opposed to the visit of the pontifex were persecuted by pro-papal state and church personnel.'

c. Nek-**e** vladike koj-**e** su bil-**e** za izbor novog
some-**F.PL** bishops who-**F.PL** AUX.PL were-**F.PL** for election new
patrijarha preko noći su promenil-**i** stav...
patriarch over night AUX.PL changed-**M.PL** attitude
'Some bishops who were in favor of electing a new patriarch changed their attitude over night....'

(31) a. A mnog-**e** vojvode su ranije bil-**i** hajduci i harambaše.
and many-**F.PL** dukes AUX.PL earlier were-**M.PL** hajduks and harambašas
'And many dukes had been hajduks and harambašas before.'

b. Čak i pravoslavn-**e** vladike, kao sv. Vasilije Ostroški, nisu se
even also Orthodox-**F.PL** bishops like st. Basil Ostrog didn't SE
ustručaval-**i** ući u veze s Rimom...
hesitate-**M.PL** get in link with Rome
'Even Orthodox bishops like Basil of Ostrog didn't hesitate to establish connections with Rome.'

(32) a. Čak su mnog-**e** srpsk-**i** 'patriote' kada više nisu
even AUX.PL many-**F.PL** Serbian-**M.PL** patriots when more didn't
znal-**e** šta da lupetaju pričal-**e** kako...
know-**F.PL** what DA blather said-**F.PL** how
'Even many Serbian 'patriots', when they no longer knew what to blather about, said how...'

b. Mogl-**i** su t-**i** velik-**e** patriote iz Dveri tada da...
could-**M.PL** AUX.PL those-**M.PL** big-**F.PL** patriots from Dveri then DA
'Those big patriots from the Dveri movement could have also...'

- Our approach handles such variability straightforwardly:
  - *Bishop* nouns have [+FEM] [+MASC] on *n*
  - Both features are copied onto agreement targets
  - Impoverishment can delete either gender feature in the plural

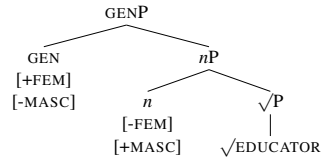
⋆ ***A note on the Agreement Hierarchy (and attempts to derive it):***

- Our examples are mostly in line with Corbett's *Agreement Hierarchy*, which states that the likelihood of semantic agreement increases monotonically from left to right in (33)

(33) *attributive — predicate — relative pronoun — personal pronoun* (Corbett 1979)

- However, they cannot straightforwardly be subsumed under analyses that derive the hierarchy effects from a two-tier gender structure (e.g., Pesetsky 2013, Matushansky 2013)

(34)



= *pedagog* 'female educator'

- (34) is an example of a hybrid noun

- A two-tier gender structure for hybrid nouns is supposed to ensure that once social gender is introduced, it becomes the target for any future agreement operations:
- If we adopted a two-tier, hybrid noun analysis for *bishop* nouns, we could not explain the pattern in (32-a), for which the following description would be necessary:[8]
  - Social gender on the (structurally lower) subsective adjective
  - Grammatical gender on the (structurally higher) intersective adjective
  - Grammatical gender on the predicate
- In contrast, an Impoverishment analysis is very much compatible with the corpus examples
  - Both gender features on *bishop* nouns are available for agreement and Impoverishment
  - It is left open which (potentially extra-grammatical) factors favor Impoverishment of one feature over the other in a given context[9]
- Thus, we have some initial evidence that *bishop* nouns are not hybrid nouns

4.3. [+MASC] *on* bishop *nouns behaves like grammatical gender, not social gender*

- Gender on Slavic nouns is generally analyzed with the two-tier structure in (34), where social gender is optional and higher on the nominal spine than grammatical gender (see e.g., Steriopolo & Wiltschko 2010, Matushansky 2013, Pesetsky 2013, Landau 2016)
- If gender agreement is governed by Agree, only the higher social gender features should be available for probing from the outside when both are present in (34)
- **Here, we demonstrate that social gender agreement behaves as expected with hybrid nouns, but not with *bishop* nouns**
- We argue this is because *bishop* nouns do not instantiate a two-tier gender structure (contra previous work, e.g., Wechsler & Zlatić 2000, 2003, Despić 2017, Puškar 2018)
- Instead, both gender features are on the same *n* head; they instantiate grammatical [+FEM][+MASC]
- BCS has two types of hybrid nouns:
  - grammatically masculine with optional feminine social gender (35)
  - grammatically feminine with optional masculine social gender (36)

(35) a. Pedagog nas je pozva-**o** na telefon. **(male or female)**
educator us AUX invited-**M.SG** on phone
'The educator called us on the phone.'

[8] This pattern can be accounted for under the idea that social gender is lower on the nominal spine than grammatical gender for *bishop* nouns, as argued in Puškar 2018. However, the data in (32-b) is problematic for such an approach.

[9] We find that linear order plays a more significant role in the mixing of gender agreement with *bishop* nouns than expected based on the Agreement hierarchy or the two-tier gender approach. While the pattern in (29) is attested regularly (F>M), the reverse pattern (M>F) does not appear in the corpus even once, despite both of them involving predicate agreement. Also, while (31) is not unusual, such patterns basically disappear if the subject is postverbal.

b. Pedagog nas je pozva-**la** na telefon. **(female only)**
educator us AUX called-**F.SG** on phone
'The educator called us on the phone.'

(36) a. Pijanica nas je pozva-**la** na telefon. **(male or female)**
drunk us AUX invited-**F.SG** on phone
'The drunk called us on the phone.'

b. Pijanica nas je pozva-**o** na telefon. **(male only)**
drunk us AUX called-**M.SG** on phone
'The drunk called us on the phone.'

- Agreement works as expected, assuming that only the (b) examples are specified for social gender–have GENP on our analysis (see Matushansky 2013 for discussion)
- A way to force social gender agreement in BCS is to use gender-specific epithets (37)

(37) a. Pedagog, žena u tridesetim, pozva-**la/*-o** nas je na telefon.
educator woman in thirties, called-**F.SG/*M.SG** us AUX on phone
'The educator, a woman in (her) thirties, called us on the phone.'

b. Pijanica, muškarac sa ulice, pozva-**o/*-la** nas je na telefon.
drunk man from street called-**M.SG/*F.SG** us AUX on phone
'The drunk, a man from the street, called us on the phone.'

- The agreement pattern in (37) is not due to the gender of the noun inside the epithet, cf. (38)

(38) Moja prijatelj-ica, strašni genijal-ac, zadivi-**la/-*o** je sve prisutne.
my friend-F.SG epic genius-M.SG impressed-**F.SG/*M.SG** AUX all present
'My female friend, an epic genius, impressed everyone present.'

- Crucially, the same pattern obtains for hybrid nouns in the plural:

(39) a. Pedagozi, žene u tridesetim, pozval-**e/*-i** su nas na telefon.
educators women in thirties, called-**F.PL/*M.PL** AUX.PL us on phone
'The educators, women in (their) thirties, called us on the phone.'

b. Pijanice, muškarci sa ulice, pozval-**i/*-e** su nas na telefon.
drunk men from street called-**M.PL/*F.PL** AUX.PL us on phone
'The drunks, men from the street, called us on the phone.'

- We do not observe the same with plural *bishop* nouns; agreement is still optionally feminine:

(40) Vladike, starci u osamdesetim, došl-**e/-i** su u sukob.
bishops old-men in eighties came-**F.PL/M.PL** AUX.PL in conflict
'The bishops, old men in (their) eighties, came into conflict.'

- This pattern is predicted if both genders on *bishop* nouns instantiate grammatical gender on *n*, but not if [+MASC] is social gender

4.4. *Bishop* nouns in BCS do not require special treatment

- Previous approaches to *bishop* nouns in BCS are (we argue) less successful than the Impoverishment analysis in certain ways
  - Puškar 2018 requires that grammatical gender be higher than social gender in the nominal spine, *pace* almost all previous work (see e.g., Steriopolo & Wiltschko 2010, Pesetsky 2013, Matushansky 2013)
  - Other approaches do not explain why *bishop* nouns do not behave like more familiar hybrid nouns, as shown in Sections 4.2 and 4.3 (e.g., Despić 2017)
- The Impoverishment analysis developed here addresses these issues
  - It relies on familiar, uncontroversial assumptions (e.g., feminine is marked relative to masculine, languages neutralize gender distinctions in the plural)
  - It predicts that *bishop* nouns do not behave like hybrid nouns because they do not have social gender features
- The Impoverishment analysis also paves the way for a unified treatment of *bishop* nouns with similar gender agreement patterns in other languages:
  - In some languages, certain nouns seem to trigger agreement in one gender in the singular and another in the plural
    - neuter nouns in Romanian (Farkas 1990, Matushansky 2022, many more)
    - certain human-gender nouns in Guébie (Kru; Sande 2019)
  - Insofar as such changes have no effects on interpretation, they can be productively modeled using Impoverishment, as in Matushansky 2022 and Kramer & Sande 2023
  - Future work: a more detailed comparison of these patterns

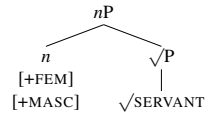
5. Cross-Slavic predictions

- The [+MASC][+FEM] analysis of *bishop* nouns extends to analogous nouns in (at least some) Slavic languages
- Extending in two directions:
  - *bishop* nouns in Russian: evidence for underlying [+MASC][+FEM] (Section 5.1)
  - *bishop* nouns in Slovenian and Old Church Slavonic: confirming the predictions of an Impoverishment approach (Section 5.2)

5.1. *Bishop* nouns are [+MASC][+FEM] in Russian

- Russian has a set of nouns cognate to *bishop* nouns in BCS (e.g., *muščina* 'man', *papa* 'dad', *dʲadʲa* 'uncle', *deduška* 'grandfather', *sluga* 'servant', *junoša* 'young man')
- We propose that these nouns have [+MASC] and [+FEM] gender features, like in BCS

(41)



= *sluga* 'servant'

- Like in BCS, they trigger obligatory masculine agreement in the singular, (42), so we propose they are subject to a similar markedness-targeting Impoverishment operation, (43)

(42) Ét-**ot** star-**yj** muščina prišël pozdno.
this-**M.SG** old-**M.SG** man came.**M** late
'This old man came late.' (adapted from Privizentseva 2023: 803)

(43) **Markedness-Targeting Impoverishment: *bishop* Nouns in Russian**
a. SD: an Agr node specified as [+MASC][+FEM]
b. SC: Delete [+FEM] in Agr

- However, Russian agreement markers do not distinguish between genders in the plural (Corbett 1991:132, Bobaljik 2001, Matushansky 2025)[10,11]
- Analysis: an Impoverishment operation deletes the gender features on plural agreement targets (Bobaljik 2001), similar to Coptic in Section 3.3

(44) **Markedness-Triggered Impoverishment for Gender in Russian**
a. SD: an Agr node specified as [+PL]
b. SC: Delete [+/-FEM],[+/-MASC] in Agr

- So, plural *bishop* nouns in Russian are predicted to trigger the same agreement as all other nouns, which is correct (data adapted from the Russian National Corpus)

(45) a. Ét-**i** muščíny ničego ne zabyl-**i**.
these-**PL.NOM** men nothing NEG forgotten-**PL**
'These men have not forgotten anything.'

[10]The same is true of Bulgarian and Macedonian, which also have cognates of *bishop* nouns that, to the best of our knowledge, behave the same as in Russian.

[11]There is one exception: *ob-a/e* 'both-M/F' displays gender agreement in non-nominative cases (see e.g., Pesetsky 2013, Privizentseva 2023). Feel free to ask a question about this.

b. Ét-**i** ženščiny byl-**i** v krinolinax ili v fižmah.
these-**PL.NOM** women.F were-**PL** in crinolines or in panniers
'These women were wearing crinolines or panniers.'

c. Ét-**i** oficery sidel-**i** pered nim, kak škol'niki,...
these-**PL.NOM** officers.M sat-**PL** in-front him like schoolchildren
'These officers sat in front of him like schoolchildren...'

d. Ét-**i** suščestva podyhal-**i** v getto, ...
these-**PL.NOM** beings.N died-**PL** in ghetto
'These creatures were dying in the ghetto, ...'

- However, under this analysis, *bishop* nouns in Russian are surface-indistinguishable from masculine nouns:
  - They trigger masculine agreement in the singular, (42)
  - They trigger the same agreement as any other noun in the plural, (45)
- If *bishop* nouns have [+FEM] underlyingly, they are predicted to pattern with feminine nouns in some surface-detectable way (at least for learnability purposes)
- Prediction borne out: declension class assignment

(46) *Russian noun classes* (adapted from Privizentseva 2023 and Alexiadou & Müller 2008)

Class	Example	Ending	Gender
I	*zavod*-ø 'factory', *student*-ø 'student', *žitel* 'inhabitant'	*-ø*	M
II	*učitel'nic-a* 'fem. teacher', *komnat-a* 'room' *muščin-a* 'man', *slug-a* 'servant'	*-a*	F
III	*tetrad'*-ø 'notebook', *doč*-ø 'daughter'	*-ø*	F
IV	*mest-o* 'place', *jablok-o* 'apple'	*-o*, *-e*	N

- Despite triggering masculine singular agreement, *bishop* nouns in Russian belong to declension class II, which is associated with feminine nouns
- Most previous work on Russian declension class treats Class II membership as dependent on having a feminine gender feature:
  - Either Class II is assigned based on a feminine gender feature (e.g., Aronoff 1994)
  - Or the Class II exponent is the realization of (in part) a feminine gender feature (e.g., Privizentseva 2023, Matushansky 2025)
- Under either analysis, Class II membership is evidence that *bishop* nouns have a [+FEM] feature = support for our analysis

- Sidebar: to the best of our knowledge, few analyses have tried to explain exactly how Russian *bishop* nouns trigger masculine agreement while also being in feminine Class II
  - Crucial recent exception: Privizentseva 2023 (P23)

(47) **P23 Analysis of Russian *bishop* nouns**

a. Syntactically masculine: [+MASC]**[-FEM]** = masc singular agreement

b. A postsyntactic morphological rule inserts [+FEM] on *n*, contextually conditioned by the roots of *bishop* nouns

c. Since the inserted [+FEM] is more marked than the underlying [-FEM], [+FEM] replaces [-FEM]

d. At Vocabulary Insertion, the presence of [+FEM] leads to the insertion of Class II exponents[12]

- There are a few reasons to prefer an Impoverishment approach to *bishop* nouns:
- It has better empirical coverage:
  - P23 predicts that *bishop* nouns will always trigger the same agreement pattern as masculine nouns
  - Therefore, it does not predict the variable gender agreement of BCS plural nouns
  - In contrast, our approach covers *bishop* nouns in both BCS and Russian
- Under P23, the gap in the typology of gender features remains puzzling (why no [+MASC],[+FEM]?)
- P23 requires a mechanism that significantly weakens the restrictiveness of DM as a theory
  - In some approaches, Impoverishment not only deletes a marked feature but also inserts the unmarked form of the same feature via a redundancy rule (see e.g., Noyer 1998)
  - However, it is assumed to be impossible to replace an unmarked value of a feature with its marked counterpart (e.g., Noyer 1998, Hein & Murphy 2025:398–399, Kramer et al. to appear)
  - But this kind of operation is crucial to P23: replacing a syntactic [-FEM] feature with a [+FEM] feature, (47-c)
  - The inability to change a feature to its marked value is a way in which DM is more restrictive than other morphological theories (e.g., Paradigm Function Morphology, Kramer 2016)
  - So, an Impoverishment account of *bishop* nouns in Rusian is preferred in that it accounts for the data **without** losing restrictiveness and using any new mechanisms
- Overall in this section, we have argued that Russian has *bishop* nouns and that an Impoverishment analysis works well to explain their behavior

[12]This is a simplification of Privizentseva's careful and through analysis of declension class and gender in Russian. See Privizentseva 2023 for details.

5.2. *Bishop* nouns in Slovenian and in Old Church Slavonic: evidence for Impoverishment

- Impoverishment operations are crucial for our analysis of *bishop* nouns:
  - Ensure masc agr in the singular: (13) for BCS, (43) for Russian
  - Ensure masc/fem agr in the plural in BCS: (22)
  - Ensure gender neutralization in the plural in Russian: (44)
- Impoverishment is motivated by markedness
  - It either deletes a marked feature directly (markedness-targeting), or deletes some other feature in the context of a marked feature (markedness-triggered)
  - A marked feature is a necessary condition for an Impoverishment operation to be posited by a learner (see e.g., Bale et al. 2011) [13]
- In this section, we confirm an additional restriction on Impoverishment (cf. Nevins 2011):

(48) **Restriction on Markedness-Triggered Impoverishment Operations**
If an Impoverishment operation is triggered by the marked feature (bundle) M, then it is triggered by the feature (bundle) M' that is more marked than M.

–The level of markedness for a feature or feature bundle can be assessed by any preferred means (e.g., node counting, markedness statements)

–Both features must be of the same kind, e.g., number, tense, etc.

- (48) is relevant for the markedness-triggered Impoverishment rule for BCS *bishop* nouns

(49) **Markedness-Triggered Impoverishment for Agr with *bishop* Nouns**

a. SD: an Agr node specified as [+MASC][+FEM][+PL]

b. SC: Delete [+FEM] or [+MASC] in Agr

- By hypothesis, (49) is triggered by the markedness of plural number
  - Dual number is more marked than plural number (see e.g., Nevins 2011)
  - **Prediction: *bishop* nouns should trigger variable agreement in the dual**
  - BCS does not have dual number, but **the prediction is borne out in Slovenian and Old Church Slavonic**:
    - These languages have *bishop* nouns (masc sg, masc/fem pl) and dual number
    - In both languages, *bishop* nouns trigger variable agreement in the dual (in addition to the plural)
- *bishop* nouns in Slovenian (data without a citation are from the web corpus SlWaC):

[13]A marked feature is not a **sufficient** condition for Impoverishment, though. For example, languages vary in whether the combination of gender and plural causes any/all gender features to be Impoverished (Russian, Coptic: yes; BCS, Classical Arabic: no)

– Slovenian has a set of nouns cognate to *bishop* nouns in BCS (e.g., *sluga* 'servant,' *starešina* 'elder,' *pismonoša* 'letter carrier', *vodja* 'leader', ...)

– Like in BCS, these nouns trigger obligatory masculine agreement in the singular

(50) Njegov-**ø/*-a** sluga je prihajal-**ø/*-a** na delo...
his-**M.SG/*F.SG** servant is came-**M.SG/*F.SG** to work
'His servant came to work...' (Nika Jurov, p. c.)

– Also like in BCS, they trigger variable masculine/feminine agreement in the plural

(51) T-**e** kolaborantsk-**e** sluge so si mislil-**i**, da bodo
those-**F.PL** collaborationist-**F.PL** servants are REFL thought-**M.PL** that would
s tem rešil-**i** sebe.
with that save-**M.PL** themselves
'Those collaborationist lackeys thought they would save themselves with that.'

(52) Starešine so bil-**e** najprej informiran-**e** ustno, nato so te iste
elders are were-**F.PL** first informed-**F.PL** verbally then are those same
informacije prebral-**i** še v vojaških glasilih, nazadnje pa so o
information read-**M.PL** also in military publications finally then are about
tem gledal-**i** še razne video prispevke.
that watched-**M.PL** also video contributions
'Senior officers were first informed verbally, then read the same information in military publications, and finally watched video reports on the subject.'

– Lastly again like in BCS, these nouns are members of a declension class associated with feminine nouns (see e.g., Priestly 1993, Börjesson 2006)

– Implications:

* The data from Slovenian show that variable agreement with plural *bishop* nouns is not a BCS-specific phenomenon
* Since the patterns are largely the same as BCS, *bishop* nouns can likely be accounted for in Slovenian using the Impoverishment analysis developed here

- Slovenian has dual number, and it is more marked than plural (Toporišič 2000, Derganc 2003, Nevins 2011, Marušič & Žaucer 2021)
- As per the prediction in (48), dual also triggers variable agreement with *bishop* nouns (data from SlWaC):

(53) a. ... in ob-**e** starešin-i sta povedal-**i** drugačno zgodbo.
. and both-**F.DU** elder-DU.NOM AUX.DU told-**F.DU** different story
'...and both elders told a different story.'

b. Ob-**a** starešin-i sta vstal-**a** sredi ljudstva in
both-**M.DU** elder-DU.NOM AUX.DU rose-**M.DU** among people and
položil-**a** roke na njeno glavo.
laid-**M.DU** hands on her head

'Both elders rose in the midst of the people and laid hands on her head.'

- We assume the following number features in Slovenian (cf. Classical Arabic, Section 3.2):

(54) a. [-PL][+MIN] = singular
b. [+PL][-MIN] = plural
c. [+PL][+MIN] = dual

- We also assume that the [+] value is marked; the [-] value is unmarked
- Therefore, plural has one marked feature and dual has two = dual is more marked than plural

Impoverishment rules for *bishop* nouns for Slovenian:

(55) **Markedness-Targeting Impoverishment: Slovenian**
a. SD: an Agr node specified as [+MASC][+FEM]
b. SC: Delete [+FEM] in Agr

(56) **Markedness-Triggered Impoverishment: Slovenian**
a. SD: an Agr node specified as [+MASC][+FEM]**[+PL]**
b. SC: Delete [+FEM] or [+MASC] in Agr

One could imagine an Impoverishment operation that only operated over the plural in Slovenian

- The SD would have the number features [+PL][-MIN]
- But such an operation is prevented from occurring because of (48), repeated as (57)

(57) **Restriction on Markedness-Triggered Impoverishment Operations**
If an Impoverishment operation is triggered by the marked feature (bundle) M, then it is triggered by the feature (bundle) M' that is more marked than M.

- (57) is a way of making more precise the relationship between Impoverishment and markedness such that falsifiable predictions can be made

Old Church Slavonic (OCS; data adapted from Huntley 1989, 1993) has *bishop* nouns

- OCS has a set of nouns cognate to *bishop* nouns in BCS (e.g., *sluga* 'servant,' *sokačьji* 'cook', *vojevoda* 'general', *vladyka* 'lord', ...)
- Singular *bishop* nouns are attested only with masculine agreement (Huntley 1993:135)
- Plural *bishop* nouns have variable masculine or feminine agreement

– *Servant* in plural, modifiers masculine

(58) služe ... boga pokrivъša i puѣtъša
servants god covering.PTCP.**M.PL** and honoring.PTCP.**M.PL**
'servants... covering God and honoring him' (Suprasliensis 458.6-8)

- *Servant* in plural, prenominal modifiers fem, postnominal modifier masc

  (59) ljutъiję i bezmilostivъiję slugъi džavolovъi
  fierce.**F.PL** and merciless.**F.PL** servants Devil.**M.PL**
  'the fierce and merciless servants of the devil' (Suprasliensis 64.3-5)

- *Cook* in plural, predicative adjective feminine, relative pronoun masculine

  (60) sokačiję čistiěšę vasъ sǫtъ iže sǫtъ rabi člověčъsti
  cooks cleaner.**F.PL** you are who.**M.PL** are servants men
  'Cooks who are servants of men are cleaner than you.' (Suprasliensis 116.2)

- Implications similar to Slovenian:
  - The data from OCS show that variable agreement with plural *bishop* nouns is not a Slovenian/BCS-specific phenomenon
  - Since the patterns are largely the same as BCS, *bishop* nouns can likely be accounted for in Slovenian using the Impoverishment analysis developed here
- OCS has dual number (Lunt 1955/2001, Huntley 1989, 1993)
- OCS *bishop* nouns are predicted to show variable agreement in dual, and this is borne out
  - *Servant* in dual, pronoun masculine, nominal modifier feminine:

  (61) va oba služe sotonině
  you.DU both.**M.DU** servant.DU Satan.ADJ.**F.DU**
  'you both are servants of Satan' (Suprasliensis 75.1)

- We propose that OCS has similar Impoverishment rules to Slovenian (omitted for space)
- This section overall: support for, and predicting the scope, of Impoverishment
  - The behavior of *bishop* nouns in OCS and Slovenian is exactly as predicted by the Impoverishment analysis
  - Specifically, (48) is borne out in that Slovenian and OCS show variable agreement with *bishop* nouns in both the plural and dual
  - Empirical predictions:
    * Metasyncretism of a feature in the plural (e.g., gender) predicts metasyncretism of that feature in the dual (TBD whether borne out cross-linguistically!)
    * Metasyncretism of a feature in the dual does not necessarily predict metasyncretism of that feature in the plural (likely borne out; e.g., Nevins, 2011, 431)
    * These predictions help differentiate morphological theories with few to no predictions about syncretism from theories like DM (see e..g, discussion in Hein & Murphy 2025)
- Interim summary: data from Russian, Slovenian, and OCS provides additional support for the Impoverishment approach to *bishop* nouns

6. Conclusion

- Our proposal has three planks:
  - *bishop* nouns have [+MASC] [+FEM] gender features
  - Markedness-targeting Impoverishment deletes the (very marked) [+FEM] feature on singular agreement targets
  - Markedness-triggered Impoverishment deletes either gender feature on plural agreement targets
- This proposal not only captures the previously-challenging agreement patterns but also:
  - Correctly models intra-speaker variation in agreement with *bishop* nouns
  - Correctly predicts declension class for *bishop* nouns
  - Correctly predicts coordination resolution patterns for *bishop* nouns
  - Is compatible with the evidence showing that *bishop* nouns are not hybrid nouns
- Taking these results together, (we argue that) an Impoverishment analysis is more successful than previous approaches to *bishop* nouns.
- Additional theoretical implications include:
  - Evidence about the nature of gender features in three-gender languages - filling in the mysterious gap
  - Support for Impoverishment approaches to changes in gender agreement across number (Kramer & Teferra 2020, Matushansky 2022, Kramer & Sande 2023)
  - Evidence that agreement resolution precedes Impoverishment
- *bishop* nouns in Russian, Slovenian and OCS:
  - Their agreement patterns are easy to capture under an Impoverishment approach, without making DM less restrictive
  - They bear out the predictions of a markedness-grounded Impoverishment approach wrt dual number
- Many open questions remain:
  - Why are all the *bishop* nouns human-denoting?
  - Are there examples in other languages of markedness-triggered Impoverishment deleting either of two features?
  - What are the implications of this analysis for languages with four genders (e.g., Tsez)?

7. References

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